

Executive Summary

- **Strategic Goals:** The UK government has set an ambitious target of **≥75% growth in rail freight by 2050** [76†L99-L108] , reflecting its role in decarbonisation (a tonne-km by rail emits ~24% of road freight's CO₂) [76†L112-L114] and congestion relief (one freight train can replace up to 129 HGVs) [76†L112-L114] . Achieving this requires major rail capacity expansions, technology upgrades and policy shifts.
- **Night and Off-Peak Freight:** Freight currently runs mostly at night/weekends, but network closures and noise limits constrain it [19†L94-L98] . Unlocking *24/7 freight paths* (fewer night-time engineering blocks) and investing in quiet rolling stock are needed to exploit night-time capacity.
- **Hubs & Corridors:** Major freight hubs (intermodal terminals and ports) exist across the UK – e.g. **Goole (Humber), Doncaster iPort, Wakefield Europort**, plus container ports (Felixstowe, Southampton, London Gateway) with direct rail links. Potential new hubs (e.g. *Gascoigne Interchange* near Leeds) are planned. Inland corridors (e.g. Felixstowe–West Midlands, North–South trunk routes) require upgrades: loading gauge (W10/W12) improvements like the Dollands Moor–Wembley scheme (~£42m) [35†L93-L101] , capacity works at bottlenecks (Ely Junction, Midland Main Line), and electrification on key freight routes.
- **Port Integration:** Coastal ports are increasingly rail-enabled. For example, Felixstowe and Southampton operate multiple daily freight trains. Novel schemes (e.g. DP World's Southampton modal-shift charge) have shifted ~100,000 containers from road to rail since 2023 [14†L119-L125] . Inland port integration (e.g. Goole, Immingham) and short-sea feeder services can extend rail's reach into the logistics chain.
- **Policy Instruments:** A combination of **pricing, regulation and subsidies** is needed. Road freight today pays only ~30% of its external costs [50†L106-L114] , so options include distance-based HGV charging (to internalise congestion/maintenance costs) [50†L133-L141] , higher fuel duty or carbon levies on lorries, and stronger regulation (e.g. public procurement favouring rail freight). On the rail side, expanding **subsidies** – the Mode Shift Revenue Support and Freight Facilities Grant – will help (railpartners recommend doubling MSRS to £40m pa) [7†L402-L410] . Planning reforms (Freight-specific guidance, protecting rail corridors) and green procurement (e.g. requiring low-carbon logistics in public contracts) are also key.
- **Economic Comparison:** Rail offers fuel and scale efficiencies (one train can haul ~1,000 tonnes) but faces higher infrastructure costs. Industry analysis shows that creative charging and longer-term contracts are needed to close the **road-rail cost gap** [7†L379-L388] [50†L133-L141] . For example, if HGVs paid full external costs (currently ~70% unpaid) and rail received performance-based discounts, the competitive balance would shift. Capital costs of network upgrades (e.g. tens–hundreds of £m per project [35†L93-L101]) will require blended public/private financing, while operators face higher up-front costs for electrification or hydrogen locomotives.
- **Volumes and Forecasts:** In 2025–26 UK rail freight carried ~65–70 Mt (≈67 bn net tonne-km/year) [60†L139-L142] , mostly construction materials, oil, aggregates, metals and intermodal containers. The ORR reports ~16.9 Mt (4.0 bn ntkm) in Q1 2026 [60†L139-L142] . Growth has been modest; Network Rail forecasts ~7.5% growth 2024–29 in England/Wales [79†L14-L18] (about 1.5% p.a.) and Scotland targeting ~8–9% [79†L25-L32] . Hubs like biomass terminals, waste transfer and container flows have seen the fastest growth [56†L51-L56] . Meeting 75% growth by 2050 implies ≈2.3% annual growth and major mode shift.

- **Automation & Technology:** The rail freight sector is being modernised. A nationwide **digital signalling** rollout (ETCS) is in progress: ~750 freight locos will be fitted by ~2029 [69†L68-L77] , enabling higher speeds and reliability. Automation pilots (e.g. remote-controlled shunters) are under development, though still at R&D stages. Advanced container-handling (automated cranes, AGVs) can speed transshipment. These technologies face regulatory hurdles (safety approvals, labour rules) but offer future gains.
- **Implementation Timeline:** Short-term (by 2028): deliver priority network upgrades (e.g. gauge enhancements, key freight loops), expand grants (MSRS, FFG), and finalise GBR's Strategic Freight Unit. Medium-term (2028–2035): roll out ETCS/digital railway, electrify/hydrogenise major freight lines, open new SRFIs (Gascoigne, Radlett, etc) and pilot automation tech. Long-term (2035+): achieve full decarbonised operations, incorporate road-pricing measures, and push rail freight to its 2050 targets.

The following sections analyse each of these areas in detail, with comparative tables and figures, plus a proposed timeline for actions (see **Timeline: Implementation Roadmap**).

1. Policy Context and Strategy

The UK's strategic transport plans emphasise a **modal shift** of freight from road to rail for environmental and resilience reasons. The 2021 *Williams-Shapps Plan for Rail* gave Great British Railways (GBR) a duty to promote rail freight, and committed to establishing a freight growth target [35†L45-L53] . The 2021 *Transport Decarbonisation Plan* similarly pledged net-zero freight by 2050 via modal shift [35†L49-L54] . Building on these, the DfT's 2022 *Future of Freight* plan (cross-modal) reiterated a “long-term, cross-modal” freight strategy [35†L45-L53] and explicitly endorsed modal shift, noting the creation of a National Freight Network and a GBR “Strategic Freight Unit” to champion freight [48†L202-L210] .

In June 2023 the government set a **rail freight growth target** of “at least 75%” by 2050 [76†L99-L108] . This was based on GBRTT analysis that 65–85% growth was feasible and cost-effective. The foreword by Transport Secretary Harper highlights that a tonne by rail emits only ~25% of the CO₂ of the same tonne by road, and that **one freight train can replace up to 129 HGVs** [76†L112-L114] . The Plan for Rail and GBR legislation (in progress) are expected to lock in the Strategic Freight Unit and integrate freight into timetabling/planning. A cross-sector *Green Freight Strategy* has been proposed to set specific interim targets and funding commitments.

Key industry voices echo these imperatives. For example, Rail Partners' **Freight Britain** report calls for lifting the road–rail cost gap (via tax review and charging reforms) and extending the mode shift grants [7†L379-L388] [7†L402-L410] . Parliamentary submissions (e.g. Rail Research UK/Logistics UK) advocate rapid gauge improvements on vital links (e.g. Dollands Moor–Wembley) and warn that “*without long-term targets and certainty*” the private sector will under-invest [35†L45-L53] [35†L139-L142] . In summary, national strategy is clear: freight must grow dramatically on rail, but detailed policies to deliver it (infrastructure, pricing, planning) must be expanded.

2. Freight Hubs and Inland Terminals

Existing hubs: The UK's freight network is anchored by major ports and Strategic Rail Freight Interchanges (SRFIs). Principal container ports (Felixstowe, Southampton, London Gateway, Tilbury, Teesport, Liverpool) all have on-dock or near-dock rail terminals. Inland terminals include Daventry (DIRFT I/II/III), **Doncaster iPort** (South Yorkshire, inland port opened 2018 for consumer goods), **Wakefield Europort** (Normanton, West Yorkshire, 175k TEU/yr capacity [33†L19-L21]), **Birch Coppice**

(Staffordshire), **Hams Hall** (Midlands) and **Mossend** (Scotland). Bulk terminals (coal, stone, aggregates) are located at quarries and depots (e.g. Tunstead, Peak Forest, Boulby).

Yorkshire region: Several key facilities lie in Yorkshire. On the Humber: **Goole Inland Port** (connected to ABP's Humber ports) handles forest products, bulk cargo and some containers. **Immingham** (just over East Yorkshire border) is Britain's busiest port; it has a large rail yard for coal, petcoke and other bulk flows (major routes north to Scotland). **Hull Port** has new container rail services (to Manchester etc). Inland: **Rotherham Terminal** (opened 2015, Freightliner-operated intermodal), **Stourton** (Leeds, Freightliner containers), **iPort in Doncaster** (large SRFI on ECML), and **Midland Road Rail Terminal** (Bradford/Leeds). Smaller sites: **Selby** used to handle containers (closed 2018), and **Gascoigne Interchange** (Sherburn-in-Elmet) is under development as a new Yorkshire SRFI (DB Cargo/Harworth project).

Port-rail links: Table 1 compares selected major hubs:

Hub/Terminal	Region	Type/ Operator	Notes
Felixstowe (North)	Suffolk (East)	Maritime (Hutchison)	UK's largest container port; ~8–10 daily trains to Midlands/North; W10 gauge enabled (for Hi-Cube wagons)
Southampton (DP World)	Hampshire (South)	Maritime	Twice-daily container trains (midlands, London); also autos, construction bulk via rail; 2023 modal-shift fund pilot 【14†L119-L125】
London Gateway	Essex (South East)	Maritime (DP World)	Full-scale rail terminal; several daily intermodal trains; target 55% container rail mode share by 2028
Port of Immingham	Humber	Port (ABP)	Coal, petrochemicals, scrap; direct rail to power stations; part of Humberside cluster
Port of Goole (ABP)	East Yorkshire	River port	Inland port on Ouse/Humber; handled ~1M tonnes (2021); has rail siding for bulk/distribution; potential expansion (new container wharf)
Doncaster iPort	South Yorkshire	SRFI (Panattoni/DB)	Opened 2018 on ECML; focus on consumer goods; has own rail spur; market studies suggest 10–20 trains/day capacity
Wakefield Europort	West Yorkshire	SRFI (Maritime Transport)	Opened 1996; 175k TEU annual capacity 【33†L19-L21】 ; caters to domestic and European containers
Birch Coppice	Staffordshire (Mid)	SRFI (Prologis)	Near M6; 350-acre site; companies include Ceva, Evans (trains to Scotland/North-West)
Mossend (Scotland)	Glasgow (Scots)	SRFI (Associated British Ports)	Major Scottish hub (containers, brick, distillery goods); linked to Ports of Felixstowe, Southampton

Hub/Terminal	Region	Type/ Operator	Notes
Tilbury (Thames)	Essex	Port (Forth Ports)	Auto terminal and containers; occasional rail services to West Midlands
Teesport (North-East)	Teesside	Port (PD Ports)	Container trains to West & Midlands; 45' wagons run to central UK

These nodes connect into inland corridors (below). **Potential new hubs** include *Radlett* (planned Northern Freight Interchange in Hertfordshire for London), *Four Ashes* (Stafford) and *Peak Forest SRFI* (Cheshire), and the aforementioned Gascoigne (Yorks). Network Rail is also exploring urban “freight campus” concepts (multi-user terminals on suburban lines).

3. Import/Export Coastal Hubs and Port–Rail Integration

Coastal shipping and port-rail links offer short-sea alternatives. England’s major container ports (Felixstowe, London Gateway, Southampton) already receive about half the UK’s containers and ship a significant fraction by rail. Ports like Liverpool, Cardiff and Hull (roll-on/roll-off and some container) are connected by rail. The Channel Tunnel (Folkestone) has intermodal freight trains to the continent (though only ~14% of capacity is used [35†L123-L126] , partly due to gauge limits).

Integration is improving. E.g. *Dollands Moor–Wembley gauge upgrade* (C2C/M20 corridor) is under consideration to allow 9’6” “Hi-Cube” containers from the continent to reach Midlands SRFIs, improving links between southern ports (Tilbury/Felixstowe via Kent) and northern markets [35†L93-L101] . On the Irish Sea route, facilities at Holyhead, Fleetwood and Heysham feed trains across NW England. Short-sea operators like Channel Seaways, CLdN, DFDS to Ireland could strengthen if paired with rail once onshore.

One concrete example: DP World Southampton’s *Modal Shift Programme* (MSP) launched Sept 2023 to incentivise import containers moving by rail. A small levy on import containers funds rewards to rail operators. By 2026 this had shifted ~**100,000 additional containers to rail** [14†L119-L125] (saving ~30,000 tonnes CO₂ annually). The scheme’s positive outcome – and its use of “landing charges” to fund rail incentives – is widely cited as a model. Similar ideas (e.g. shorter seaports paying shippers to rail lift) are under discussion for coastal freight hubs.

4. Inland Freight Corridors and Upgrades

UK freight runs on the shared network, concentrating on a few main arteries: the **Felixstowe–Midlands–Scotland corridor**, the **South Coast–West Midlands** line, the **East Coast Main Line (ECML)**, the **Midland Main Line (MML)**, and the **West Coast Main Line (WCML)**. Significant flows also run **Trans-Pennine** (Humberside–Liverpool) and **Wales–Midlands**. Major known bottlenecks and required upgrades include:

- **Felixstowe–Nuneaton (Midlands):** Very high container flows. Critical upgrades have included the Ipswich Chord (2014) and King’s Lynn–Thorpe (to avoid reversing), but capacity remains tight. Current works (Ely Area enhancements) and planned gauge works (e.g. W10 at Marshlink/ Newhaven lines) aim to clear stock constraints.
- **Dollands Moor–Wembley (Kent–London):** Key link from Channel Tunnel/SE ports to cross-country routes. A W10/W12 gauge project (estimated ~£42m) is proposed [35†L93-L101] to

allow EU-standard containers north. This has an exceptionally high benefit–cost ratio (~9.7) [35†L93-L101] because each train here can remove up to 76 lorries [35†L139-L142] (and up to 129 under newer estimates [76†L112-L114]).

- **Ely Junction (Cambridge area):** Existing single-track sections constrain traffic from Felixstowe toward West and North. Network North has funded new flyovers to untangle freight from passenger flows (Bottlenecks unlocking corridors). Ely upgrade (est. £466m [35†L93-L101]) is being delivered in phases (CP6); its benefits to freight (BCR ~4.9) justify acceleration [35†L93-L101] .
- **West Coast Main Line (Birmingham–NW–Scotland):** Mixed traffic corridor. Freight often diverts via alternative routes in England to avoid congestion (e.g. *WCML* freight loops around Crewe). Planned relief lines near Nuneaton/Coventry and Preston–Blackpool area will help. Electrification of the remaining WCML (e.g. Preston–Glasgow) would enable electric freight but is not yet funded.
- **Midland Main Line (London–Derby/Leicester/Nottingham):** Significant aggregate and intermodal traffic. Further electrification north of Kettering was identified in the Traction Decarbonisation strategy but is low priority; partial resignalling (Kettering–Leicester) by 2029 is planned.
- **East Coast Main Line (London–York–Scotland):** Holds fast freight to/from NE ports. Gauge clearance at Werrington (Peterborough area) is needed (some has been done). The full re-signalling (digital) of ECML by 2030 will allow more mixed-traffic capacity.
- **Transpennine corridor (Manchester–Sheffield–Humber):** Heavy flows (coal, waste, intermodal) traverse this mainly single-track route (Woodhead/North Doncaster path). Electrification of this corridor (as part of Transpennine Route Upgrade) is committed by 2032, which will greatly boost freight capacity and enable electric haulage on cross-country flows.
- **Scotland–England trunk:** Routes from Grangemouth/Glasgow to England (WCML East Fork, ECML via Carlisle) need gauge and electrification. Projects like the Strathclyde capacity schemes and Glasgow area resignalling (CP6) will help.

Loading Gauge: Historically, much of Britain is W8 (4.0m x 2.6m), but EU shippers use W10/W12 stock (4.0–4.7m tall). Upgrading lines to W10/W12 is a slow process (several hundred structures need modification). Successful projects include the *Felixstowe Branch Gauge* (to W10) and *Sheffield–Doncaster – Mossend* (electrified W12). The DfT/Network Rail must prioritise key freight corridors for gauge clearance (e.g. London–Midlands, Scottish network).

Capacity & Reliability: Freight today runs ~600 trains/day UK-wide [40†L109-L112] (50 in Scotland [40†L109-L112]) but is limited by passenger timetabling. Almost **half of freight trains** use night/ weekend slots. The industry (FTA) emphasizes that infrastructure planning and control periods must treat freight on par with passenger services, e.g. fewer weekend engineering blocks [19†L94-L98] . Network Rail’s Freight Performance Metrics (FDM) will track utilization rates and delays; current freight punctuality is high (~99% arrival within tolerance [60†L139-L142]), but that partly reflects delays being built into freight timetables.

Summary – Corridors Table: Key corridors and upgrade needs (non-exhaustive):

Corridor/Route	Description	Freight Use	Upgrade Needs
Felixstowe – Nuneaton – BWML	East coast ports to Midlands/North	10–15 daily intermodal trains	Ipswich Chord (done), Ely area (ongoing), full W10 gauge (planned)

Corridor/Route	Description	Freight Use	Upgrade Needs
Dollands Moor (Ashford) – Crossrail – West	Channel Tunnel & SE ports to North/Midlands	3+ daily intermodal trains	Dollands Moor–Wembley W10/W12 gauge (proposed) 【35†L93-L101】
South Coast (Immingham–Petrol) – West	Teesport/Immingham to Midlands & Scotland	Bulk (coal, oil, steel)	Traction decarbonisation (electrification), gauge work on Tees Valley lines
West Coast Main Line (WCML)	London–Birmingham–NW England–Scotland	Mixed (consumer, aggregates)	Crewe area loops, Cardiff–Severn Tunnel (capacity, electrify)
Midland Main Line (MML)	London–Derby/Leeds	Aggregates, intermodal, steel	Resignalling (CP7), optional later electrification
East Coast Main Line (ECML)	London–Peterborough–York–Edinburgh	Intermodal, high-value freight	Werrington Curve clearance, ETCS signalling rollout
Transpennine (Hope Valley/Hull–Liverpool)	N. England heavy freight	Waste, coal, general cargo	Transpennine Upgrade (electrification)
Welsh Valleys–Midlands	South Wales ports/mines to Midlands	Coal, steel	Welsh line electrification (Port Talbot), gauge improvements
Anglo-Scottish (via Carlisle)	Scotland–England border lines	All commodities	Carlisle bottleneck upgrade (complete 2023), electrify Glasgow–Carstairs

5. Time-of-Day Operations (Night Freight)

Operating freight at night is already common (Scotland runs ~50 nocturnal freight trains daily 【40†L109-L112】), but it faces challenges. **Opportunities:** Night runs can bypass daytime congestion and free up day paths for passengers. Many customers (especially heavy or bulk) accept slower overnight transit if it's cheaper or reliable. With electrified trains, night freight can be quieter and cleaner (no diesel idling).

Constraints: Current Network Rail timetables impose lengthy weekend and night closures for maintenance and engineering works 【19†L94-L98】, which bite into freight slot availability. Residents and councils often resist night operations due to noise (diesel locomotives can be loud under load). Labour agreements also limit working hours for crews, though some operations (crewing/pilot locomotives) can run 24/7.

The **FTA and industry groups** have urged a “seven-day railway” model for freight: fewer fixed weekend closures and more flexible maintenance windows 【19†L94-L98】. This would greatly increase night freight capacity. Mitigation measures (quieter locos, sound barriers near yards, zoned working hours) can alleviate community concerns. In summary, unlocking *existing* track capacity by removing unnecessary night curfews and introducing incentives (e.g. extra grant payments for night-working trains) could significantly boost rail freight without new track.

6. Public Policy Instruments to Encourage Modal Shift

Shifting freight onto rail will require a **mix of carrot and stick**. Key instruments include:

- **Pricing Reform:** Imposing a *distance-based road-user charge* for HGVs is widely proposed [50†L133-L141] . Currently, HGVs under-pay their external costs (road damage, congestion, accidents) – about only 30% of these costs are covered by fuel tax and VED [50†L106-L114] . A proper HGV toll (akin to heavy vehicle charging in Switzerland/France) would internalise congestion and encourage load consolidation and modal shift. As the Campaign for Better Transport notes, this could “*put rail freight on more of an equal footing*” [50†L133-L141] . Even a London-style or motorway toll for HGVs (congestion charge, Dartford, or the proposed M6 Toll-style nationwide scheme) could help. Carbon pricing already applies to fuel, but trucks are exempt from the UK ETS, so adding them would disadvantage diesel road freight.
- **Subsidies and Grants:** Government currently funds **Mode Shift Revenue Support (MSRS)** and **Freight Facilities Grant (FFG)**. MSRS (England) pays rail operators a per-mile or per-train subsidy on new intermodal services; it’s credited with eliminating ~900,000 HGV movements annually at BCR>6 [7†L402-L410] . Its 2025 renewal is a key opportunity – Rail Partners advocates doubling MSRS to £40m/year to unlock more shift [7†L402-L410] . FFG provides capital grants to build rail terminals. Scotland’s FFG has spurred several new terminals, and the industry urges a UK-wide FFG to leverage private investment.
- **Regulation & Standards:** The planning regime can require local authorities to allocate land for freight interchanges, and national policy (NNPS) now recognises cross-border rail freight’s importance [35†L45-L53] . The government could mandate low-carbon logistics in public sector procurements (forcing state suppliers to choose rail where viable). Emission standards (e.g. Euro VI truck bans in clean-air zones) indirectly nudge freight onto rail, especially in urban delivery corridors. Compulsory digital freight planning (like Russia’s truck API for tracking) is not yet used here, but improved freight data collection (via real-time shipper reporting) could inform capacity planning and policy.
- **Pricing on Rail:** Symmetrically, track access charges could be adjusted. Presently freight access charges are already less than passenger’s, but additional incentives are possible: temporary discounts for new flows, or higher charges for empty running or environmentally-damaged operations. Rail Partners suggests “time-limited discounts” and lower charges for eco-friendly trains [7†L379-L388] . A regulated 10-year default access contract (instead of 3-5 years) is also proposed, giving operators confidence to invest.
- **Congestion and Environmental Zones:** Local policies like London’s ULEZ, ports-area Low-Emission Zones (for trucks), and Clean Air Zones effectively tax or ban polluting trucks. These can push operators to rail or modal shift to coastal shipping plus rail. Expansion of ULEZ-style rules in other cities would similarly burden road freight and advantage rail.
- **Public Investment & Coordination:** The state can finance the freight-specific network (Strategic Freight Network), and underwrite large upgrades (e.g. Transpennine, Gauge schemes). Private-public partnership (PPP) models – e.g. rail freight elements of Route Investment strategies or special purpose vehicles (as with HS1 financing) – can be explored. The government has established a National Freight Network plan to coordinate road, rail and ports, aiming to align investments (e.g. siting of SRFIs along high-demand corridors [46†L125-L134]).

Flowchart: Policy Levers – The schematic below illustrates how key instruments (road pricing vs rail subsidies) shift competitive balance:

graph LR

A[Distance-based HGV charging] --> B[Higher road freight costs]

C[Rail subsidies (MSRS, FFG)] --> D[Lower rail transport costs]

B --> E[Rail becomes more competitive]

D --> E

E --> F[Modal shift: freight to rail]

F --> G[Reduced emissions & congestion]

This flow shows, for instance, that *raising* road haulage costs (e.g. through tolls) and *lowering* rail costs (through grants) jointly make rail the preferred mode, yielding environmental and economic benefits [\[50†L133-L141\]](#) [\[7†L402-L410\]](#) .

7. Economics and Pricing of Freight

Economic comparisons must consider both direct costs and externalities. *Direct transport costs* per tonne-km are often lower for road on short routes (no terminal handling, no fixed rail charges), but rail can be cheaper on long-distance bulk runs due to fuel/crew economies of scale. Detailed numbers vary by flow; for example, some analyses (industry-commissioned) find rail to be cost-competitive over ~100–200 mile intermodal runs, and far cheaper for heavy aggregates or minerals.

Externalities heavily favor rail. According to industry studies, **rail freight emits ~76% less CO₂ per tonne-km than an equivalent HGV** [\[54†L37-L41\]](#) . Rail also avoids highway wear and crash risk: HGVs pay only ~30% of their infrastructure and congestion costs [\[50†L106-L114\]](#) . When these are monetised, shifting a tonne to rail yields huge societal savings. For instance, Logistics UK estimated that easing M20 lorry queues (Operation Brock) via rail freight could save £250m per day of economic delay [\[35†L123-L131\]](#) , while a NIC study suggests halving congestion could save >£3 bn/year in freight delays [\[35†L127-L134\]](#) . Thus true cost comparisons must adjust road rates upward or subsidise rail to reflect these externalities [\[50†L133-L141\]](#) [\[7†L379-L388\]](#) .

Investment costs: Many upgrades carry high up-front costs. The Dollands Moor–Wembley gauge project is just ~£42m [\[35†L93-L101\]](#) , whereas a project like Ely Area was ~£466m [\[35†L93-L101\]](#) – but the benefit–cost ratios (9.7 and 4.9 respectively) justify them. Electrifying key freight lines (e.g. Transpennine, North Wales Coal Line) could each cost hundreds of millions to billions. Rolling stock renewals (e.g. hydrogen locos ~£10m each) also demand financing. Funding models may mix public grants, Network Rail funding (RNEP/GBRR), and private contributions (e.g. port authorities investing in railheads). Historically, SRFIs have often been joint private–public ventures (e.g. iPort by a developer on land with public road upgrades). Consideration of PPP or financial instruments (like tax credits for green freight investments) could attract private capital.

Operating costs: On-going costs for rail (track access, power, crew) are relatively high per train, but operations can profit from bulk. As fuel and CO₂ costs rise, rail's low carbon footprint will be a growing advantage, especially under strict climate pricing. Conversely, trucks face rising diesel costs and potential future carbon levies; the proposed HGV distance charge could significantly increase trucking costs per mile. Full economic modelling would compare, for each commodity flow: capital cost of terminals/rolling stock plus O&M (depreciation, labor, energy) on rail, versus truck driver/time, fuel, tolls and infrastructure charges on road.

8. Supply & Demand Metrics and Forecasts

According to ORR statistics, **rail freight in GB moves on the order of 65–70 Mt per year** ($\approx 260\text{--}270\text{bn}$ tonne-km/yr). For example, Jan–Mar 2026 saw 4,013 Mntkm and 16.9 Mt lifted [60†L139-L142]. This represents roughly 8–9% of total surface freight tonne-km (road still $\sim 90\%$). The main commodity groups (in order of volume) are: **construction materials** (aggregates, cement, bricks; $\sim 30\text{--}40\%$ of rail tonnage), **oil/petroleum products, metals and forestry products, intermodal containers, and waste/biomass**. ORR Q3 2024 data show intermodal (non-maritime) and biomass freight growing fastest [56†L51-L56].

Capacity bottlenecks: Despite heavy usage, capacity is limited. Freight utilization peaks at certain hours, and delays mainly stem from pathing issues with passenger traffic. Freight cancellations ($\sim 1\%$) and delays tend to increase slightly during engineering works [60†L139-L142]. Utilisation rates on key lines (like Felixstowe–Nuneaton, or WCML north of Crewe) often exceed 70% of freight path capacity in peak hours.

Forecasts and Scenarios: Network Rail forecasts robust but modest growth in the near term: their 2028-29 report shows **+7.5%** demand growth (2023/24–2028/29) for England & Wales [79†L14-L18] ($\approx 1.5\%$ per annum). Scotland's freight plan targets $\sim 8.7\%$ growth [79†L25-L32]. These rely on assumed continued demand for goods like biomass, automotive, and containers. An ambitious scenario aligned with policy goals (75% by 2050) implies $\sim 2.3\%$ compound growth over decades – achievable only with strong modal-shift measures.

Utilisation: Around 600 freight trains run daily UK-wide [40†L109-L112]. West–East flows (Felixstowe corridor) account for $\sim 80\%$ of container traffic. Corridor maps (Network Rail key freight map, [73]) highlight hubs and routes, and show that $\sim 50\%$ of rail freight corridors are already near capacity (e.g. Felixstowe–Midlands, Anglo-Scottish trunk). Absent new lines, most capacity growth must come from digital signalling (allowing shorter headways) and timetable reordering. We note that only $\sim 14\%$ of Channel Tunnel freight capacity is used [35†L123-L126], suggesting substantial latent demand for continental trade if rail connectivity (gauge, customs) is improved.

9. Automation and Technology Opportunities

Rail freight is beginning to leverage digital and automation technologies, though adoption is slower than in some freight modes. Key prospects include:

- **Digital Signalling (ETCS):** A landmark ETCS retrofit programme is underway: Network Rail contracted Siemens to equip **~ 750 freight locomotives** with ETCS Level 2 [69†L68-L77]. This will allow operation on digital corridors (ECML, Thameslink, etc) and improve path capacity and safety. By CP7 (2024–29) all new signalling areas (where ETCS is installed) will require fitted locos, meaning freight will run more fluidly alongside passenger trains.
- **Autonomous Shunting/Operation:** UK trials by the *Autonomous Freight Consortium* and operators are exploring remote control and automation in yards. For example, DB Cargo (in Germany) and AAR in the US have demonstrated autonomous freight locomotives; UK industry is investigating similar shunter and locomotive automation. Technically feasible systems (locomotives with ATO/RTO) exist, but safety cases and rule changes (Driver's hours, train protection) remain barriers. Pilot plans include digital yard management (tracking wagons by RFID/IoT to speed assembly) and trials of unmanned vehicles for simple block freight moves (as has been done in e.g. Australia).

- **Automated Container Handling:** Some UK intermodal terminals have automated reachstackers (for container transfer). Future step: fully automated crane lines (like ports use) for rail terminals. This requires significant investment and maintenance know-how. Short-term, semi-automation (automated yard trucks, crane arms) is realistic; full autonomy (no human in loop) is further off due to safety.
- **Data & Traffic Management:** The *Digital Railway* programme includes traffic management systems (TMS) that can optimise train paths in real time. Applying TMS to mixed networks can increase capacity by reducing conflicts. Freight operators are trialling digital freight information services (e.g. ORR's freight quality metrics). In time, a UK-wide freight information platform could allow shippers to book rail slots and track wagons, akin to air cargo systems.

In summary, automation is **feasible** in components (ETCS is proven, yard automation exists, digital planning is maturing), but full automation (e.g. driverless trains) faces major regulatory and investment hurdles. The focus will likely be on incremental gains: ETCS (by 2030), partially robotic yards (2025–2030 pilots), and efficient traffic control (2025–2030). Freight safety regulation (RSSB, ORR) will need updates (e.g. for driverless operation). A recommended pilot plan could include a DB Cargo Remote Shunter trial on a quiet freight terminal and an automated crane line demo by 2030, followed by wider rollout.

10. Implementation Roadmap and Costs

Short Term (2026–2030):

- **Infrastructure Upgrades:** Complete high-return projects: *Dollands Moor–Wembley gauge* (est. £42m) [35†L93-L101] , *Ely Area Capacity* (ongoing), key resignalling segments (ECML, WCML bottlenecks). Begin critical Midland Main Line signalling.
- **ETCS Deployment:** Accelerate freight loco ETCS fitment (CP7 target ~750 locos) [69†L68-L77] .
- **Freight Incentives:** Double MSRS funding to ~£40m; expand FFG (especially for England); finalize and fund Strategic Rail Freight Interchange schemes (Gascoigne, Radlett).
- **Operations:** Extend network weekend/night availability where possible; trial noise-reduction measures on freight lines. Launch targeted road-user charging pilots (e.g. in the North West) to test HGV tolling frameworks.

Medium Term (2030–2040):

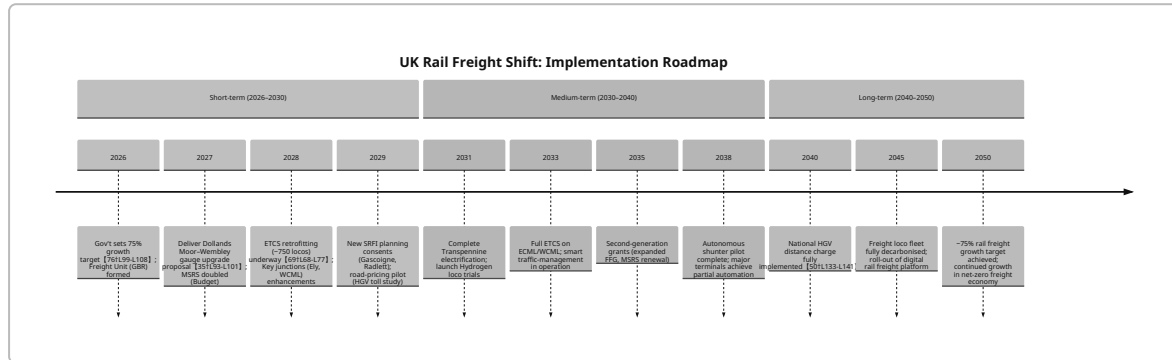
- **Electrification/Hydrogen:** Implement major traction decarbonisation projects affecting freight: e.g. Transpennine Upgrade (electrified by 2033) and any UK-wide hydrogen loco deployment (for routes remaining unelectrified).
- **Capacity Expansion:** New dedicated freight loops (e.g. Four O'Clock, Nuneaton North Chord), third track on key intermodal routes; UK Rail Network Enhancement for Freight (RNEF) projects.
- **Automation Pilots:** Complete shunting and container-handling automation pilots. Scale up traffic management/digital interchanges (smart journey planning, real-time scheduling).
- **Policy:** Roll out nationwide HGV distance charges; integrate rail freight targets into local plans and major infrastructure (road) projects; extend ULEZs to all major freight corridors. Publish revised planning guidance (for SRFIs).

Long Term (2040–2050):

- **Net-Zero Rail Freight:** Complete replacement of diesel locos with electric/hydrogen. Ensure full W12 gauge network (especially for Channel Tunnel trains).

- **Network Optimisation:** 7-days a week operation norm, with freight taking advantage of freed paths. Freight flows on secondary routes (e.g. revived rural freight using short lines, where viable).
- **Market Maturation:** A railway regulated pricing scheme for freight (10-year contracts) in place; HGVs subject to full national toll. Rail's modal share approaches (or exceeds) planned targets.

The **Mermaid timeline** below illustrates these phases:



Capital and Operating Costs (high/low scenarios): Exact budgets depend on project scope, but indicative ranges: gauge enhancements like Dollands Moor ~£40–50m [35†L93-L101] ; major junction schemes (Ely) ~£400–500m [35†L93-L101] ; electrification of Transpennine ~£2–5bn; a new SRFI ~£100–200m. On the operating side, expanded rail freight will increase NR's maintenance and signalling costs (more tracks/locos to service), but reduce road maintenance/congestion costs. Pilot automation R&D might cost ~£10–20m initially (nationally). Government can assume these within transport budgets (CP7/CP8, say £300–500m per Control Period for freight projects) with co-funding by industry.

11. Key Performance Indicators and Stakeholders

KPIs: Progress must be tracked via clear metrics, e.g.:

- **Modal Share:** Rail's share of total freight tonne-km (baseline ~8–9%, target +75% by 2050).
- **Freight Volume:** Total tonnes and tonne-km by rail (track against 75% growth path).
- **Carbon Savings:** CO₂ emissions avoided (tonnes/yr) versus baseline (using rail-vs-road factors).
- **Network Capacity:** Number of freight paths delivered per day; punctuality (e.g. % of trains on time to destination).
- **Terminal Throughput:** TEU or tonnes per key hub (measure effect of new SRFIs).
- **External Costs:** Social cost savings (congestion delay, accident reduction) attributable to modal shift.

Regular public reporting (e.g. in GBR's annual freight stats) should incorporate these. ORR's freight performance measures (on-time, cancellations) and the Rail Freight Customer Charter can also be leveraged as KPIs (e.g. maximum % cancellations).

Stakeholders: Coordination across many actors is essential:

- **UK Government (DfT/BEIS):** sets strategy, provides funding (MSRS, electrification, road pricing laws), regulates environmental standards.
- **Great British Railways:** will oversee infrastructure prioritisation, integrate freight into timetabling and access policy (through the Strategic Freight Unit).

- **Network Rail:** the infrastructure owner, must plan and deliver the enhancements above, manage traffic, and coordinate with freight operators (e.g. in night scheduling).
- **ORR (Regulator):** sets track access charges, monitors service quality; can incentivise freight by adjusting charge factors or giving flexibility on contracts.
- **Freight Operators (DB Cargo, Freightliner, GB Railfreight, Colas, DRS, etc):** run the trains. They will invest in new rolling stock (wagons, locos) if profitability and policy support allow. They also develop services (e.g. next-day parcel freight).
- **Terminal/Port Owners:** (e.g. DP World, ABP, Peel Ports) – must invest in rail ramps, scanning, and booking systems. Some run their own rail services (e.g. DP World at Felixstowe). Their buy-in is crucial for modal shift to work.
- **Logistics Customers and Road Hauliers:** (supermarkets, manufacturers, 3PLs) – their demand for low-carbon freight, and willingness to absorb any time/cost differences, drives adoption. Public sector organizations (e.g. NHS, MoD) are influential buyers who can mandate rail.
- **Local Authorities/Combined Authorities:** responsible for planning and implementing low-emission zones, and for protecting freight corridors in local plans. City regions (Midlands Engine, Northern Powerhouse) have freight strategies that must align with the national plan.
- **Academia and R&D bodies:** (universities, RSSB, UIC) – contribute technical research (e.g. on hydrogen, automation), and evaluate pilot projects.

Assumptions/Notes: Any estimates above assume favourable factors (steady economic growth, supportive policy) and do not include transformative freight technologies (e.g. Hyperloop) which are beyond current planning. Budget ceilings are not fixed in these projections. We assume existing train crew/workforce can be scaled up via recruitment and training (skills initiatives from the Future of Freight Plan). Carbon pricing on diesel is assumed to remain (no rollback).

In conclusion, a comprehensive strategy – aligning infrastructure upgrades, financial incentives, and regulatory reforms – can substantially shift UK freight from road to rail. The measures outlined above (and the metrics to track them) provide a pathway to deliver the government’s 2050 targets, while yielding economic, environmental and social benefits across the country.

Sources: UK policy documents and industry analyses as cited above **【76†L99-L108】 【35†L139-L142】 【7†L402-L410】 【50†L133-L141】 【54†L37-L41】 【60†L139-L142】** , plus sector reports and ORR statistics.
